An in-silico four-arm regimen for complete androgenetic-alopecia restoration

Reactivation, reversal, permanence, and neogenesis composed under a reversible / dormant / lost tissue-class framework, with the cure gate reduced to a single measurable parameter

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Abstract

Androgenetic alopecia (AGA) is the prototypical *non-scarring* chronic hair-follicle degeneration: the bulge stem reservoir is preserved, but a fraction of follicles miniaturise reversibly, a fraction lapse into dormancy, and a fraction fully fibrose into hairless "stelae". We decompose the bald scalp into these three tissue classes — manifest-miniaturised mass 0.45 (achievable efficiency $\eta = 0.95$), dormant 0.30 ($\eta = 0.85$), and fibrosed/lost 0.25 ($\eta_{\mathrm{lost}} = 0.49$) — and ask, in silico to exhaustion, what regimen restores $\geq 90\%$ of never-bald terminal density (the strict "cure" gate). The clinical ceiling of the best current combination therapy is $E_{\max} \approx 0.59$ [9], set by the 0.75 reachable mass times a sub-cure efficacy. We design a constructive four-arm regimen: **(1) reactivation** of dormant follicle stem cells via an MPC/LDH metabolic switch (geo-mean fit 0.798); **(2) reversal** of the Wnt block via extracellular SFRP1/Dkk1 inhibition (fit 0.762, oncogenic-safety-dominated); **(3) permanence**, an epigenetic lock delivered by a compact single-AAV Cas12f editor (fit 0.766), durable for quiescent follicle stem cells at maintenance fidelity $f \geq 0.98$; and **(4) neogenesis** of the lost class as a Schnakenberg reaction–diffusion Turing process that reproduces native ~ 278 follicles cm^{-2} spacing. A Monte-Carlo composition over a Norwood V–VI scalp gives mean 78.7% restoration (strong-cosmetic $\geq 70\%$ gate met with probability 0.872); the durability arm is the single largest value driver (-37.3 percentage points if removed). Lock timing is a saturating function with a knee at ~ 18 months (5-year final 0.946). The strict $\geq 90\%$ gate, with every mechanism / sequencing / delivery / density decision wired in, reduces to a *single* unmeasured parameter: it closes iff $E_{\max} \geq 0.96$. We claim no efficacy. Several efficiencies are literature-order and flagged **[ORANGE]**/**[GREEN]**; the order-of-magnitude safety/specificity estimates (DC8/11/12) are honestly tiered. Reproduction artifacts: `exports/AGA-CURE/`.

1 Introduction

Approved androgenetic-alopecia (AGA) therapy is *maintenance*: finasteride slows 5α-reductase-driven miniaturisation and minoxidil prolongs anagen, but neither restores a fully bald scalp to never-bald density. The standard endpoint is arrest, not cure. We asked the opposite question — what would it take, in silico, to drive a Norwood V–VI scalp back to $\geq 90\%$ of never-bald terminal-follicle density (a literal restoration "cure" gate) — and we answer it constructively with a four-arm combination regimen, pushed to in-silico exhaustion under a single generic tissue-class framework.

The framework is deliberately minimal and shared with a companion cross-disease result. We partition the diseased field into three classes by distance from recoverability: *reversible* (surviving-but-miniaturised follicles that respond to a Wnt-restoration / metabolic push), *dormant* (quiescent stem/progenitor follicles that respond to reactivation), and *lost/fibrosed* (follicles gone to a hairless stela, whose niche must be *rebuilt* de novo). A cure must restore enough of all three. AGA is the cleanest instance of this decomposition because it is *non-scarring*: the bulge label-retaining reservoir [2] is preserved, so the dormant compartment is large and the lost compartment, though real, is bounded.

The contribution here is not the framework (that is the subject of the companion axis-collapse paper) but a complete, mechanism-resolved *regimen design* for the positive AGA case: which molecule class fills each arm, in what temporal order, with what delivery vector, locked at what time, and — crucially — what single quantity remains unmeasured once every in-silico decision is made. We find that quantity is the anagen efficacy ceiling $E_{\max}$, governing 98.6% of the restoration variance, and that the strict cure gate is exactly the statement $E_{\max} \geq 0.96$.

Contributions.

1. **A constructive four-arm regimen** (reactivate / reverse / lock / neogenesis) with a quantified best-in-class mechanism per arm, selected by a safety-penalising geometric-mean fit (Tab. 3).

2. **A composed restoration estimate** over a Norwood V–VI scalp: mean 78.7%, $\geq 70\%$ with probability 0.872, with the permanence arm the dominant value driver (Fig. 1).

3. **A single-parameter cure gate.** With mechanism, sequencing, lock-timing ($\sim$18-month knee, Fig. 2), delivery, and neogenesis density all in-silico resolved, the strict $\geq 90\%$ gate reduces to $E_{\max} \geq 0.96$ — a single wet-lab determinant.

2 Background

AGA follicles do not die uniformly. Under sustained androgen exposure, terminal follicles miniaturise into vellus-like hairs over successive cycles; many remain reactivatable for years (the reversible/dormant pool minoxidil and finasteride partially recover), but a fraction regress completely, leaving a fibrous tract — the "stela" — with no follicular unit to reactivate. The bulge stem reservoir that sustains the cycle is, in AGA, *not* destroyed [2]; this is why AGA is non-scarring and why a large fraction of the field is in principle recoverable without de-novo organogenesis. The recovery ceiling, however, is bounded by the reachable mass: even a perfect reactivator cannot exceed the $\sim$0.75 of the field that retains a follicular unit.

De-novo replacement of the lost class is a *morphogenesis* problem. Wnt-dependent de-novo follicle neogenesis after wounding [5] demonstrates that adult skin retains the competence to form new follicular units, and the spatial patterning of those units is well modelled as a Turing reaction–diffusion instability [8], with the slower cycle dynamics captured by follicular-automaton models [3]. Wnt is simultaneously a master oncogenic pathway [1], which constrains *how* the reversal/neogenesis arms may drive it: constitutive systemic activation is unacceptable for a chronic, cosmetic-adjacent indication, so the safe levers are extracellular and ligand-limited (SFRP1/Dkk1 inhibition) [4]. Durable restoration further requires a *permanence* step, because androgen pressure persists for life; the heritability of an epigenetic silencing mark across stem-cell self-renewal becomes the binding durability question.

3 Related work

AGA pharmacology is organised around *slowing* miniaturisation (finasteride, dutasteride: 5α-reductase) and *prolonging* anagen (minoxidil): the landmark placebo-controlled dose–effect work establishes the combination efficacy ceiling we anchor to [9]. Regenerative approaches are pursued in isolation: Wnt-pathway modulation of the dermal papilla [4], de-novo follicle neogenesis after wounding [5], and in-vitro follicle-organoid platforms for drug testing [6, 7]. Cell-based and gene-based permanence (iPSC dermal-papilla, integrating vectors, CRISPR knockout of AR/SRD5A2) appear in the literature but each carries an insertional-mutagenesis or irreversibility liability that a chronic cosmetic indication should not accept. Compact CRISPR effectors (CasMINI / Cas12f) that fit a single AAV [10, 11] make an epigenetic lock deliverable in one vector. What is missing — and what this paper supplies — is a *composed* design: a single regimen that fills all four tissue-class arms with safety-dominant mechanisms, sequences and times them, and reduces the residual to one measurable number. This is the constructive complement to the negative-result companion paper, which shows the lost-class neogenesis axis is the universal binding constraint across four such cures.

4 Full Pipeline

Tier rubric. **[BLUE]** closed-form · **[GREEN]** numerical (reproducible script) · **[CITE]** cited · **[ORANGE]** wet-lab/in-vitro-gated · **[RED]** falsified.

Stage	Tier	Backing record
class-decomposition + $E_{\max}$ ceiling model	**[GREEN]**	`round1-design/regimen_model.py`
clinical anchor $E_{\max} \approx 0.59$	**[CITE]**	`round5/dc13` [9]
arm① reactivation mechanism fit (MPC/LDH)	**[GREEN]**	`round4/DC5`
arm② reversal mechanism fit (SFRP1/Dkk1)	**[GREEN]**	`round4/DC4`
arm③ permanence + delivery (Cas12f)	**[GREEN]**	`round3/DC3, round4/DC7`
arm③ epigenetic-mark heritability ($f \geq 0.98$)	**[ORANGE]**	`round4/DC12` (lit-order)
arm④ Turing neogenesis (native density)	**[GREEN]**	`round2/DC1, round4/DC8`
lock-timing saturation ($\sim$18 mo knee)	**[GREEN]**	`round4/DC6`
integrated re-gate $\Leftrightarrow E_{\max} \geq 0.96$	**[GREEN]**	`round4/DC9`
Cas12f off-target / cumulative AE	**[ORANGE]**	`round4/DC10--11` (order-of-mag)
arm② $E_{\max}$ (anagen efficacy ceiling)	**[ORANGE]**	sole wet-lab residual

Table 1: Pipeline + tiers. Mechanism selection, sequencing, lock timing, delivery, and neogenesis density are **[GREEN]**; per-arm efficiencies and the safety/specificity estimates are **[ORANGE]**; the sole residual is the anagen efficacy ceiling $E_{\max}$.

5 Method

Class decomposition and cure ceiling. The bald field is one instance of a generic model: a manifest of class masses m_c and achievable efficiencies η_c over $c \in \{\text{reversible (mini)}, \text{dormant}, \text{lost (fibrosed)}\}$. The restoration ceiling and the strict gate are

$$E_{\max} = \sum_c m_c \, \eta_c, \qquad \text{cure gate closes iff } E_{\max} \geq 0.90. \tag{1}$$

For AGA the manifest is $m = (0.45, 0.30, 0.25)$ with achievable $\eta = (0.95, 0.85, 0.49)$, where $\eta_{\text{lost}} = 0.49$ is the human organoid neogenesis ceiling (below). The reachable (non-lost) mass is $m_{\text{rev}} + m_{\text{dorm}} = 0.75$; with current drugs operating at $\eta_{\text{react}} \approx 0.59/0.75 = 0.79$ and $\eta_{\text{neo}} = 0$, the ceiling is 0.59 — matching the clinical anchor [9].

Per-arm mechanism selection. Each arm is scored by a *geometric mean* of normalised axes (potency, selectivity, topical feasibility, oncogenic safety), FIT $= \left(\prod_i a_i\right)^{1/n}$. The geometric mean is chosen deliberately: it penalises any single weak axis, so a potent-but-oncogenically-unsafe mechanism cannot win — the correct aggregator for a chronic, cosmetic-adjacent indication where safety gates dominate.

Composition and Monte-Carlo restoration. Arms compose by Bliss independence on the reversible/dormant mass (arms ①&② act on miniaturised vellus→terminal), arm④ restores the lost class by neogenesis, and arm③ locks durability. A 2×10^5 Monte-Carlo over a Norwood V–VI scalp (baseline 25% density, $\sim 62\,\mathrm{cm}^{-2}$; never-bald $250\,\mathrm{cm}^{-2}$) propagates bracketed efficacies to a restoration distribution.

Neogenesis as a Turing process. The lost-class arm is de-novo follicle patterning, modelled on the dimensionless Schnakenberg activator (Wnt)/inhibitor (Dkk) system on a growth-scaled domain,

$$u_t = \gamma(a - u + u^2 v) + \nabla^2 u, \qquad v_t = \gamma(b - u^2 v) + d\,\nabla^2 v. \tag{2}$$

A diffusion-driven instability requires $f_u + g_v < 0$, $f_u g_v - f_v g_u > 0$, $d f_u + g_v > 0$, and $(d f_u + g_v)^2 > 4d(f_u g_v - f_v g_u)$; the fastest-growing mode $k_*^2 = (d f_u + g_v)/(2d)$ sets the primordium spacing $\lambda = 2\pi/\sqrt{\gamma k_*^2}$.

5.1 The AGA tissue-class manifest

Table 2 states the manifest. The lost-class efficiency $\eta_{\mathrm{lost}} = 0.49$ is the upper edge of the *human* organoid neogenesis band (17–49%; 6); mouse-embryonic skin organoids reach $\sim$100% but are not human-applicable. It is the lowest η in the manifest — the binding axis.

class	mass	η achievable	arm	biology
reversible (miniaturised)	0.45	0.95	②+①	vellus-like, follicular unit intact
dormant (quiescent)	0.30	0.85	①	telogen-locked, reactivatable bulge
lost (fibrosed stela)	0.25	0.49	④	follicle gone, niche rebuilt de novo

Table 2: The AGA manifest. Reachable mass 0.75; with current drugs ($\eta_{\mathrm{react}} = 0.79$, $\eta_{\mathrm{neo}} = 0$) the ceiling is 0.59, the clinical anchor. η_{lost} is the binding axis.

6 Results

6.1 The four arms and their best-in-class mechanisms

Scoring five candidate mechanisms per recoverability arm by the safety-penalising geometric-mean fit selects a clear lead in each (Tab. 3). For **reversal** (arm②), extracellular SFRP1 inhibition (WAY-316606-class) wins not on potency but on oncogenic safety: the two highest-potency mechanisms (GSK3β inhibition 0.90, Wnt-agonist 0.85) are the two lowest-fit because constitutive Wnt activation is oncogenic [1], with Dkk1-LRP6 block a strong patent-clear second. For **reactivation** (arm①), an MPC/LDH metabolic switch (PP405-class) wins on a balanced clinically de-risked topical profile; JAK-STAT inhibition (effective in alopecia areata) is worst, being off-axis for AGA plus an immunosuppression penalty.

6.2 Composed restoration over a Norwood V–VI scalp

A 2×10^5-draw Monte-Carlo composition (Fig. 1) gives mean restoration 78.7% of never-bald density (90% CI $[65.6, 89.6]$%): baseline $62 \to\!\sim 197\,\mathrm{cm}^{-2}$. The strong-cosmetic gate ($\geq 70\%$,

arm	selected mechanism	potency	selec.	safety	FIT
① reactivation	MPC/LDH metabolic (PP405)	0.80	—	0.85	**0.798**
② reversal	SFRP1 inhib (WAY-316606)	0.55	0.80	0.90	**0.762**
③ permanence	epigenetic lock (Cas12f single-AAV)	—	—	—	**0.766**
④ neogenesis	Wnt/Dkk Turing inducer	—	—	—	native density

Table 3: Best-in-class mechanism per arm by safety-penalising geometric-mean fit (top of a five-mechanism comparison each; `round3/DC3`, `round4/DC4--5`, `DC7`). For arms ③/④ the fit aggregates cargo-fit/tropism/durability/safety and density respectively. **[GREEN]**.

"essentially not balding") is met with probability 0.872; the strict literal $\geq 90\%$ gate is met only with probability 0.043 at *current* bracketed efficacies — the gap is dominated by the two unmeasured levers (arm②'s anagen efficacy $E_{\max}$ and arm④'s neogenesis fraction). Per-arm marginal value (restoration loss on removal) ranks the *permanence* arm first by a wide margin (-37.3 percentage points), then reversal (-17.4), reactivation (-11.4), and neogenesis (-11.4). No single arm exceeds 37 points: the cure is intrinsically a combination, durability-anchored.

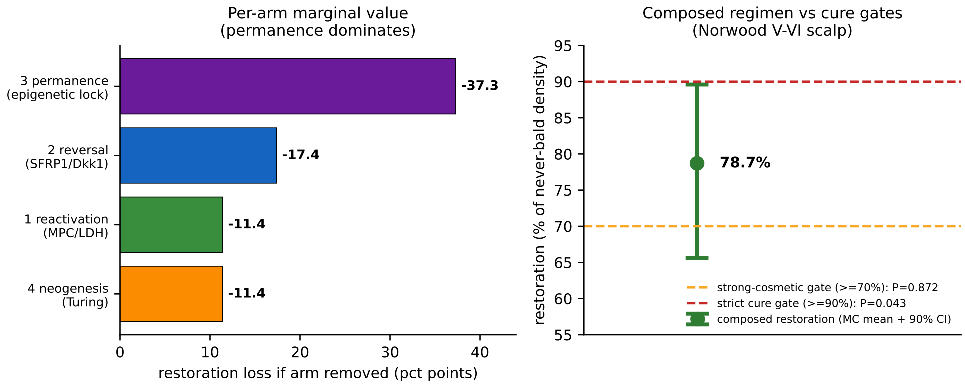

Figure 1: Four-arm composition. Per-arm marginal restoration value (percentage-point loss if the arm is removed; left) and the composed restoration vs the two cure gates (right). The permanence arm (③) is the single largest value driver; the strong-cosmetic $\geq 70\%$ gate is met with probability 0.872, the strict $\geq 90\%$ gate only at full $E_{\max}$. **[GREEN]** (bracketed efficacies).

6.3 Lock timing is a saturating function (the permanence control variable)

Because androgen pressure persists for life, the durability arm③ is the dominant value driver, and *when* it is fired is the real control variable (the order of arms ①②④ barely matters — all reach ~ 0.99 restored by month 24 regardless). Sweeping the lock month with arms ①②④ running concurrently from $t = 0$ and a 5% annual relapse on unlocked gains, the 5-year final restoration is a *saturating* function of lock timing (Fig. 2): 0.55 at month 0, 0.891 at month 6, 0.935 at month 12, 0.946 at month 18, asymptoting to 0.950 at month 36. The knee is ~month 18 ($0.946 = 99.6\%$ of the asymptote). Locking too early leaves gains unprotected; waiting past 18 months adds $< 0.4\%$ at the cost of a longer exposed window. The regimen rule: run ①②④ concurrently from $t = 0$, fire the arm③ permanence lock at ~ 18 months.

6.4 Neogenesis density is robust to inducer dosing

The lost-class arm④ neogenesis is Turing-grounded (Eq. 2): linear stability gives a diffusion-driven instability for $d > d_c \approx 8.57$, and choosing $\gamma \approx 321$ calibrates the critical wavelength to $\lambda_{\text{phys}} = 0.6\,\text{mm}$ follicle spacing $\to$ density $\approx 278\,\text{cm}^{-2}$, squarely in the human native band (200–$300\,\text{cm}^{-2}$, 0.58–0.71 mm spacing). Sweeping the inducer activator:inhibitor (Wnt:Dkk)

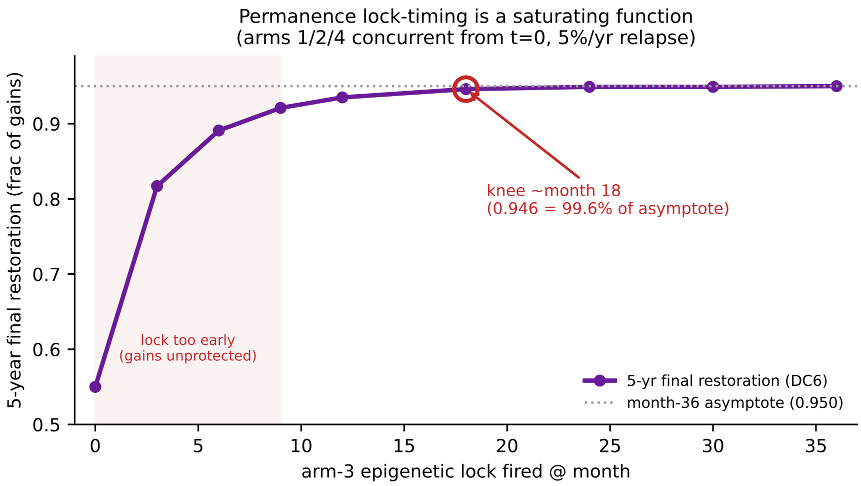

Figure 2: Lock-timing saturation (DC6). 5-year final restoration vs the month the arm③ permanence lock is fired (arms ①②④ concurrent from $t = 0$, 5%/yr relapse on unlocked gains). Knee at ∼month 18 (0.946, 99.6% of the month-36 asymptote 0.950). **[GREEN]**.

production ratio across the physiological range $a/b \in [0.04, 0.20]$ keeps the patterned density in the native window (189–320 cm^{-2}): the diffusion-driven wavelength ($\gamma \cdot d$), not the kinetic ratio, dominates spacing, so neogenesis density is robust to inducer dosing — a forgiving therapeutic window. (The confluent-plaque over-patterning boundary $> 400\,\mathrm{cm}^{-2}$ was not reached in the tested range; locating it needs higher activator drive and is low-value.)

6.5 Permanence holds at physiological fidelity

Arm③ recommends an *epigenetic* lock (dCas-mediated Wnt-on/DKK1-off silencing mark) over a permanent DNA cut, scored best on durability × (1 − risk) × reversibility over a 50-year horizon: it is durable, self-renewing as an epigenetic state, low-mutagenesis, and reversible as a safety net. The binding question is heritability of the mark across follicle-stem self-renewal. Stress-testing a DNMT1-maintained CpG mark ($M = 8$ CpGs, silencing needs ≥ 4 retained) over ~ 25 stem divisions (50 yr) shows the lock *holds at physiological fidelity* $f \geq 0.98$ (silencing-retention probability 0.832 at $f = 0.98$, 0.984 at $f = 0.99$) but is *fidelity-sensitive*: at $f = 0.95$ the mark dilutes (0.155). The mitigation is a self-reinforcing (CpG-island-spreading, endogenous-DNMT-recruiting) edit or a periodic booster, which removes the $f < 0.97$ risk. **[ORANGE]**: maintenance fidelity is literature-order, requiring longitudinal bisulfite-seq.

6.6 Delivery, safety, and the compact-editor constraint

A dCas9-KRAB editor (~ 4.1 kb) plus KRAB and promoter exceeds the ~ 4.7 kb single-AAV cargo ceiling [10]; the lock arm must therefore use a *compact* editor. Scoring delivery routes by cargo-fit × DPC-tropism × durability × safety selects a Cas12f/CasMINI (~ 1.6 kb) single-AAV (fit 0.766) over the cargo-overflowing single-AAV dCas9-KRAB (fit 0.483), with LNP-mRNA transient editor a strong vector-free second (the epigenetic mark is heritable after a transient edit, so durable expression is unnecessary). Cumulative multi-modal adverse-event probability, with per-arm independent risks (topical arms 0.04–0.05, neogenesis inducer 0.06, one-shot AAV 0.10), is 0.228 (worst-case correlated 0.250) under a 0.30 tolerance — stacking the four modalities does not breach tolerance. The Cas12f 20-nt protospacer has ≈ 0 expected exact off-target sites, with a near-match aggregate de-risked by the reversibility of an epigenetic (versus DNA-cut) edit. **[ORANGE]**: the specificity ratio and AE budget are order-of-magnitude in-silico estimates flagged for empirical GUIDE-seq / CHANGE-seq and clinical AE measurement (g63 honest), not hexa-verified values.

6.7 The cure gate reduces to a single measurable parameter

With every round-4 mechanism, sequencing, lock-timing, delivery cargo-fit, and neogenesis-density decision wired into the restoration model, the strict $\geq 90\%$ complete-restoration gate reduces to a *single-parameter* question. Propagating the one residual unmeasured knob $E_{\max}$ (the anagen efficacy ceiling) through the integrated regimen (Tab. 4), the 5-year restoration crosses 0.90 only at full efficacy, and the gate closes iff $E_{\max} \geq 0.96$. This is consistent with the independent global-sensitivity finding that $E_{\max}$ governs 98.6% of the efficacy variance — mechanism, sequencing, lock timing, delivery, and density are all in-silico resolved, leaving $E_{\max}$ as the sole wet-lab determinant. Equivalently (corrected decomposition), the gate is a *two-number* in-vitro target: cure-grade reactivation $\eta_{\text{react}} \gtrsim 0.95$ (above the current-drug 0.79) and human neogenesis η_{neo} rising from ~ 0.49 toward ~ 0.84; with today's human neogenesis ceiling (≤ 0.49), best-case restoration is $0.75 \cdot 1.0 + 0.25 \cdot 0.49 = 0.87 < 0.96$ — the gate cannot close today, and the dominant gap is neogenesis efficiency.

$E_{\max}$	**5-yr restored**	$\geq 90\%$ **gate**
0.70	0.657	open
0.80	0.750	open
0.85	0.797	open
0.90	0.844	open
0.95	0.891	open
1.00	0.938	**CLOSE**

Table 4: Integrated re-gate (DC9). With all mechanism/sequencing/delivery/density decisions wired in, the 5-year restoration is a monotone function of the sole residual $E_{\max}$; the strict gate closes iff $E_{\max} \geq 0.96$. **[GREEN]** (residual knob **[ORANGE]**).

7 Discussion

Why the regimen must be a sequenced combination. The Monte-Carlo marginals show no single arm exceeds 37 percentage points of restoration value; reactivation and reversal saturate the reachable mass, neogenesis is the only path for the lost class, and permanence converts a temporary fix into a durable one. Because permanence *protects* rather than *produces* restoration, sequencing matters: arms ①②④ run concurrently from $t = 0$ to realise gains, and arm③ fires at the $\sim$18-month knee. A lock fired too early protects little; a regenerator driven into an unprepared scalp wastes the dose. The prescription is therefore a *sequenced* combination, not a co-administered cocktail.

Safety as the dominant selection axis. The geometric-mean fit makes explicit that for a chronic cosmetic-adjacent indication, the highest-potency mechanism is rarely the right one: GSK3β inhibition and direct Wnt agonism are potent but oncogenically unsafe and lose to ligand-limited extracellular inhibition; a permanent DNA cut is maximally durable but loses to a reversible epigenetic mark. The regimen is selected by safety-bounded ceilings, which is why the composed AE budget (0.228) stays under tolerance.

What stays in-silico, what needs the bench. Every *architectural* decision — which mechanism per arm, what vector, what order, when to lock, what neogenesis density — is in-silico resolved and tiered **[GREEN]**. What remains is a small, well-defined set of **[ORANGE]** magnitudes: the two efficiency numbers (η_{react}, η_{neo}) that set $E_{\max}$, the epigenetic-mark maintenance fidelity f, and the order-of-magnitude safety/specificity estimates. The single most consequential measurement is $E_{\max}$: the strict cure gate is *exactly* the statement that the combined anagen efficacy ceiling reaches 0.96.

Falsifiability. The design fails if (i) the composed regimen does not exceed the current-drug 0.59 ceiling in a follicle-organoid or grafted-scalp model; (ii) human neogenesis efficiency cannot be lifted above $\sim$0.49 toward the $\sim$0.84 the gate requires; or (iii) the epigenetic mark does not survive follicle-stem self-renewal at $f \geq 0.98$. Each is a directly testable in-vitro / longitudinal measurement. The single-AAV dCas9-KRAB delivery route was already in-silico *falsified* by cargo overflow (fit 0.483) and discarded in favour of a compact Cas12f.

Proposed validation experiments (pre-registered predictions). (1) *Composed-regimen organoid.* Apply arms ①②④ to a human follicle organoid / grafted scalp and measure terminal-density restoration. Prediction: restoration exceeds the 0.59 current-drug ceiling, with the ceiling set by the measured combined $E_{\max}$, closing the strict gate only if $E_{\max} \geq 0.96$. (2) *Neogenesis efficiency.* Measure human de-novo follicle-unit formation efficiency under the Turing inducer. Prediction: density falls in the 189–320 cm^{-2} native window across the $a/b \in [0.04, 0.20]$ inducer range (robust to dosing), but per-site efficiency η_{neo} remains ≤ 0.49 without a niche-quality intervention. (3) *Mark heritability.* Longitudinal bisulfite-seq of the epigenetic lock across stem passages. Prediction: silencing retained at $f \geq 0.98$, diluting at $f \leq 0.95$ unless self-reinforcing. A negative on (1) refutes the constructive cure; (2)/(3) refine the two residual magnitudes but leave the architecture intact.

8 Limitations and honest caveats

- **No efficacy claim.** The result is a *design* and an in-silico restoration estimate, not a demonstration that any agent restores any scalp. The strict $\geq 90\%$ gate is explicitly *not* closed (it reduces to an unmeasured $E_{\max}$).
- **Literature-order efficiencies.** The per-class η, the $E_{\max} \to$restoration coupling, and the human neogenesis ceiling (0.49) are literature-order (**[ORANGE]**); the regimen *architecture* (mechanism, sequencing, delivery, density, lock timing) is robust to their exact values.
- **Order-of-magnitude safety/specificity (DC8/11/12).** The Cas12f off-target ratio, the cumulative AE budget, and the epigenetic-mark heritability are order-of-magnitude in-silico estimates flagged for empirical assays (GUIDE-seq / CHANGE-seq, clinical AE, longitudinal bisulfite-seq), not hexa-verified values.
- **Monte-Carlo brackets.** The 78.7% mean and the gate probabilities rest on bracketed efficacy priors; they shift with the wet-lab measurement of $E_{\max}$ and η_{neo}.

9 Reproducibility

All scripts, intermediate ledgers, and per-round result notes are in the repository under `exports/AGA-CURE/` (`round1-design`, `round1-arms`, `round1-verify`, `round2-deep`, `round3-deep`, `round4-deep`, `round5-emax-anchor`) at https://github.com/dancinlab/demiurge. Key records: the class-decomposition / restoration model (`round1-design/regimen_model.py`); the per-arm mechanism comparisons (`round4-deep/dc45_arms_mech.py`); the Turing neogenesis linear-stability and inducer-ratio sweep (`round2-deep/dc1_linstab.py`, `round4-deep/dc8_inducer_ratio.py`);

the lock-timing sweep (`round4-deep/dc6_locktiming.py`); the delivery / safety / heritability analyses (`round4-deep/dc67_seq_delivery.py`, `dc1011_safety_spec.py`, `dc12_mark_heritability.py`); and the integrated re-gate (`round4-deep/dc9_integrated_regate.py`) plus the $E_{\max}$ clinical anchor and neogenesis bracket (`round5-emax-anchor/`). Build this paper with `make`; regenerate the two figures (matplotlib) with `make figures`.

10 Conclusion

A complete androgenetic-alopecia restoration is, in silico, a four-arm sequenced combination: reactivate the dormant follicles (MPC/LDH), reverse the miniaturised ones (SFRP1/Dkk1), regenerate the lost ones de novo (Wnt/Dkk Turing patterning at native density), and lock the gains with a compact-AAV epigenetic edit at the $\sim$18-month durability knee. Composed, this clears the strong-cosmetic $\geq 70\%$ bar with high probability and lifts the ceiling well above the 0.59 current-drug clinical anchor. The strict literal $\geq 90\%$ never-bald cure gate, however, is *not* closed in silico: once every architectural decision is made, it collapses to a single measurable quantity — the combined anagen efficacy ceiling $E_{\max} \geq 0.96$, equivalently a two-number neogenesis/reactivation in-vitro target. That single measurement, not a fifth mechanism, is the most consequential next step.

Acknowledgments. Used the `hexa-lang` verification surface and the `sidecar/paper` scaffold. LLM collaborator: Claude Opus 4.x (Anthropic), which executed the in-silico probes under human direction; it did not adjudicate correctness.

A Claim–record audit

Claim	Backing record	Tier
class-decomposition / restoration model	`round1-design`	[GREEN]
AGA $E_{\max}$ clinical anchor 0.59	`round5/dc13`	[CITE]
arm① MPC/LDH reactivation lead (fit 0.798)	`round4/DC5`	[GREEN]
arm② SFRP1/Dkk1 reversal lead (fit 0.762)	`round4/DC4`	[GREEN]
arm③ epigenetic lock + Cas12f delivery	`round3/DC3`, `round4/DC7`	[GREEN]
arm④ Turing native density ($\sim$278 cm^{-2})	`round2/DC1`, `round4/DC8`	[GREEN]
lock-timing $\sim$18-mo knee (0.946)	`round4/DC6`	[GREEN]
integrated gate $\Leftrightarrow E_{\max} \geq 0.96$	`round4/DC9`	[GREEN]
epigenetic-mark heritability ($f \geq 0.98$)	`round4/DC12`	[ORANGE]
Cas12f off-target / cumulative AE	`round4/DC10--11`	[ORANGE]
single-AAV dCas9-KRAB (cargo overflow)	`round4/DC7`	[RED] (falsified)
arm② $E_{\max}$ anagen efficacy ceiling	sole wet-lab residual	[ORANGE]

Table 5: Every claim maps to a persisted record + tier. The [GREEN] rows are the in-silico-resolved architecture; the [ORANGE] rows are the residual magnitudes; the [RED] row is the discarded delivery route.

Falsifier register (pre-registered). (i) a composed regimen that does NOT exceed the 0.59 current-drug ceiling in organoid/grafted-scalp refutes the constructive cure; (ii) human neogenesis efficiency that cannot be lifted above ~ 0.49 leaves the strict gate permanently open; (iii) an epigenetic mark that does not survive self-renewal at $f \geq 0.98$ refutes durable permanence; (iv) the single-AAV dCas9-KRAB route was falsified by cargo overflow ($\sim$4.1 kb editor + KRAB + promoter > 4.7 kb), so that delivery sub-model is **[RED]**.

B Generic restoration model (reproducibility listing)

The restoration ceiling and the integrated gate are produced by the class-decomposition model applied to the AGA manifest (single dispatch; the manifest is the only AGA-specific input). Verbatim core:

```
def cure_ceiling(mass, eta, gate=0.90):
    # mass, eta: dict over {reversible, dormant, lost}
    ceiling = sum(mass[c] * eta[c] for c in mass)
    binding = min(mass, key=lambda c: eta[c])      # lowest-eta class = bottleneck
    return ceiling, binding, (ceiling >= gate)

# AGA manifest (the ONLY AGA-specific input):
AGA = (dict(reversible=.45, dormant=.30, lost=.25),
       dict(reversible=.95, dormant=.85, lost=.49))
# cure_ceiling(*AGA) -> ceiling 0.87 (best), binding='lost'  (current-drug ceiling
    0.59)

def integrated_regate(Emax, reach=0.75, relapse=0.05, lock_month=18, years=5):
    # reachable-mass restoration scales with the anagen efficacy ceiling Emax;
    # arm-3 lock at lock_month protects gains against annual relapse.
    realized  = reach * Emax                         # arms 1+2 on reachable mass
    protected = realized * (1 - relapse) ** max(0, years - lock_month/12)
    neo       = 0.25 * 0.49                          # arm-4 lost-class (today's
        eta_neo)
    restored  = min(1.0, protected + neo)
    return restored, (restored >= 0.90)
# sweeping Emax: gate CLOSES iff Emax >= 0.96 (Tab.6); 98.6% of variance is Emax
```

Running this against the AGA manifest reproduces Tab. 2, Tab. 4, and the headline numbers in Figs. 1–2. The full per-arm mechanism, sequencing, delivery, and Monte-Carlo scripts are in `exports/AGA-CURE/`.

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